



Accelerating Inference

Matthew Povey – SambaNova EMEA
ISC 2026 - Hamburg



The SambaNova Journey

Snapshot:

- Founded in 2017 by AI pioneers with roots at Stanford University
- 400+ employees across Palo Alto, Austin, London, Bangalore, Tokyo, Singapore, and Australia
- \$1.4B+ funding raised through Series E

Founders



Rodrigo Liang
Co-founder & CEO



Kunle Olukotun
Co-founder & Chief Technologist & Stanford Professor

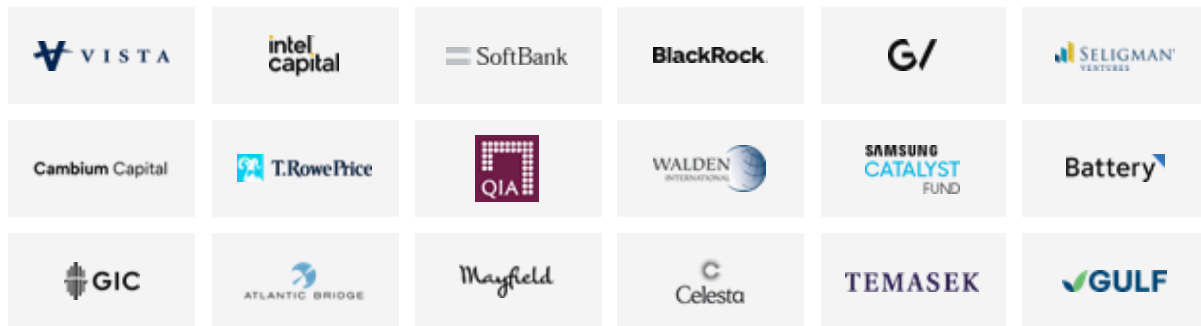


Christopher Ré
Co-founder & Stanford Professor



Lip-Bu Tan
Chairman

Sophisticated, long-term investors and strategic new investors

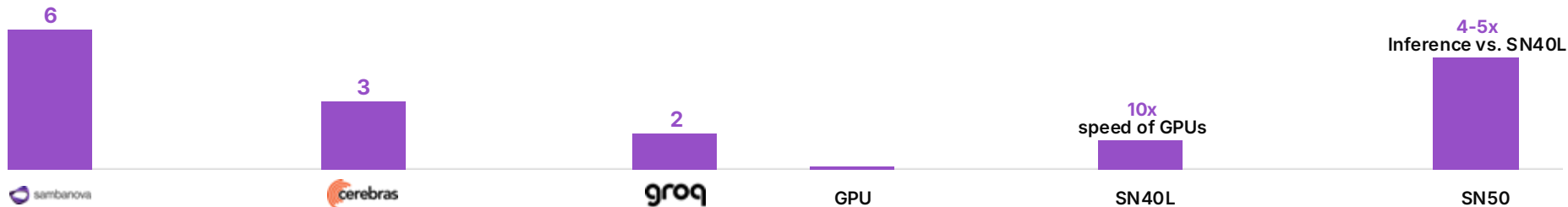


Five Generations of Dataflow Innovation

From SN10 to SN50, SambaNova has continued to advance its inference architecture across multiple chip generations.



Tape-Outs Last Seven Years



Superior Chip Performance for AI Inference

Premium Inference is the new Standard

"**NVIDIA** just admitted the general-purpose GPU era is ending"

Matt Marshall

January 3, 2026

1-10

Tokens

Entering the Trillion Parameter Era

25+

Hours

Agents Are Running for Days instead of Hours on current GPUs

250+

Token/s/user

Enterprises are paying a premium for speed on intelligent models



Dylan Patel

@dylan522p

Open models show 2.5x faster, 6x more expensive

Lower batch size, speculative decoding harder

Pareto optimal curve for Deepseek at InferenceMAX.ai shows this Claude Opus 4.6 is 100 Tok/s/user

Deepseek at 100 is 6k Tok/s/GPU

At 250 tok/s/user it's closer to 1k

Long Horizon Coding Agent on OpenAI Codex

Using GPT-5.3-codex at Extra High reasoning

ACTIVE RUNTIME

25.4h

29.7h wall-clock session span

TOKEN USAGE

12.7M

10.7M input • 2.0M output

TOOL CALLS

2,106

1,391 shell calls • 632 patches

PATCH VOLUME

26.0k

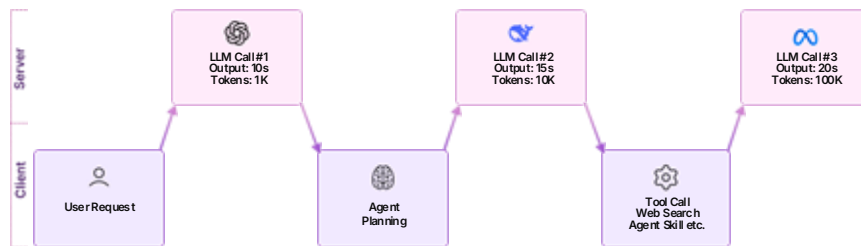
Appropriate lines added

PEAK SINGLE STEP

270,352

Tokens in one turn

Long Horizon Coding Agent on OpenAI Codex



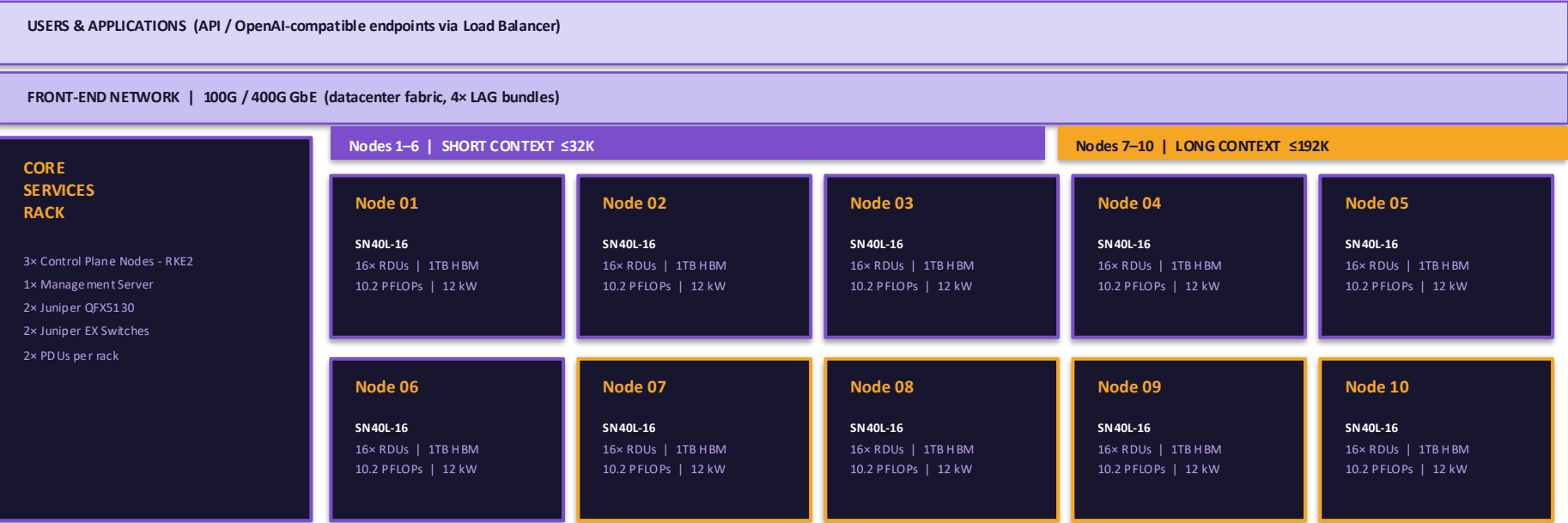
01. End-to-end latency

02. Cost per task

03. Task success/quality

Latency compound across every step of the workflow

10-Node Starter Cluster: System Architecture



Management Network: 1 GbE (OOB) | Back-end RDU Private Network | Customer-Provided: DNS • NTP • LDAP/OIDC • 15TB+ NFS Storage

SN40L: SambaNova's 4th Gen AI Chip

Reconfigurable Dataflow Unit (RDU)

Native multi-tenancy support with fast model switching

Ideal for production inference, multi-tenancy, agentic workflows

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SN40L RDU



● 3-tier Dataflow Memory

● 520 MB On-Chip
SRAM Memory



Very fast memory for high speed inference with caching

● 64 GB High
Bandwidth Memory



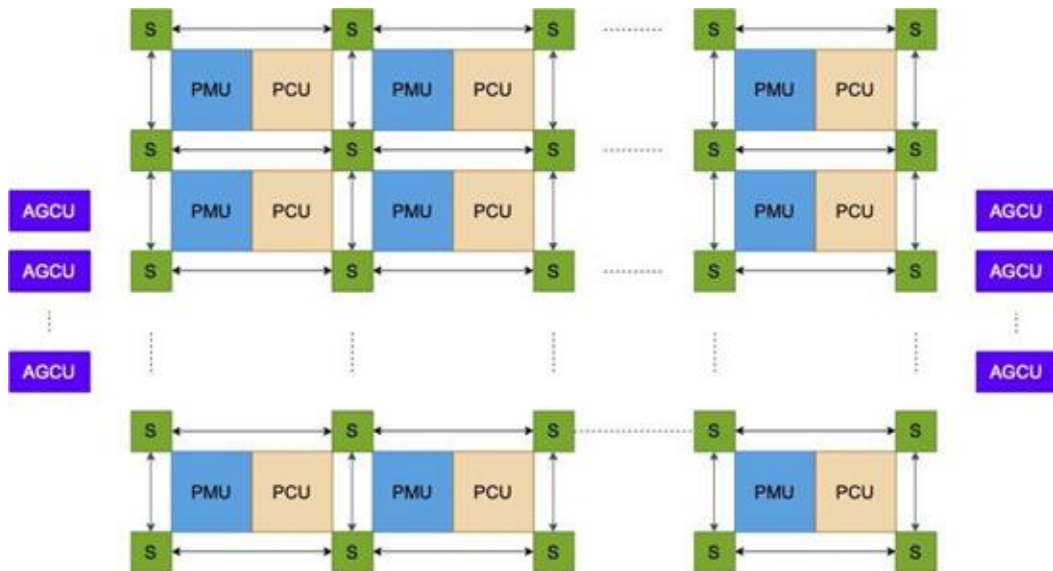
Switch between models in milliseconds

● 768 GB High Capacity
DDR Memory



Hold large number of models in memory

SN40L: Tile Architecture



1040 PCUs and PMUs

PCU: Compute unit

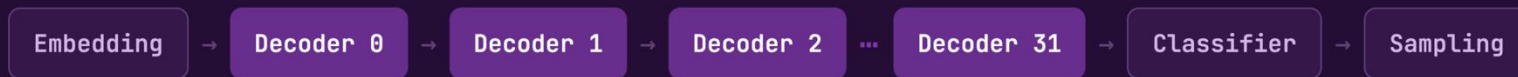
PMU: Memory unit

S: Mesh switches

AGCU: Portal to off-chip
memory and IO

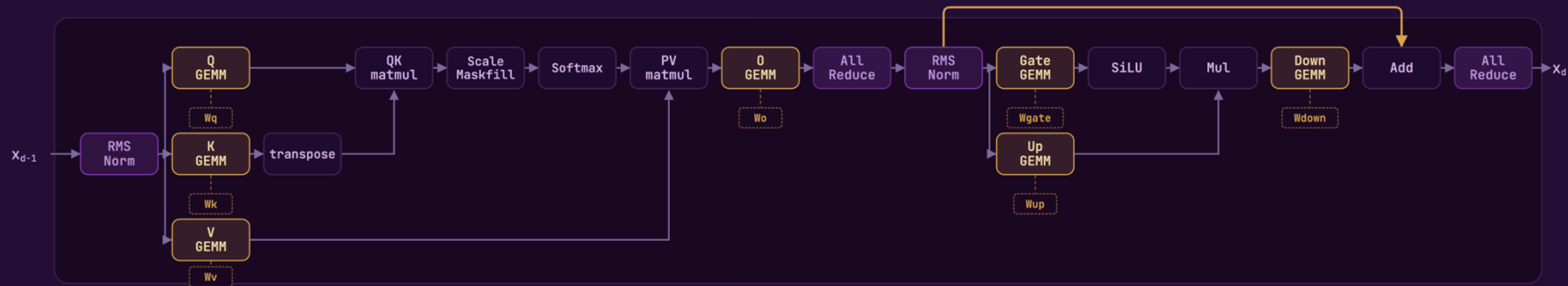
EXAMPLE · LLAMA 3.1 8B

Structure of a Transformer

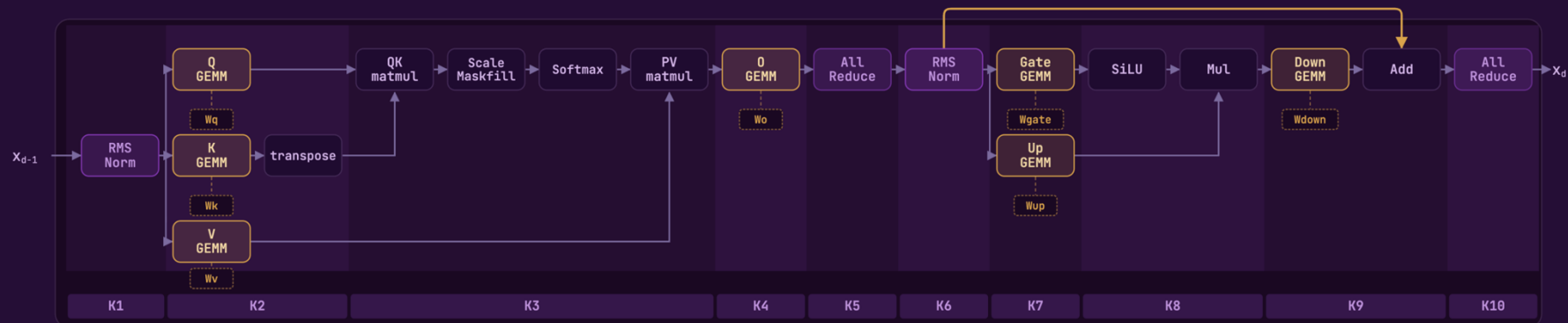


32 identical decoder layers — the dataflow graph inside each one drives everything that follows.

Structure of a Transformer



GPUs Incur Overhead with Low Operator Fusion

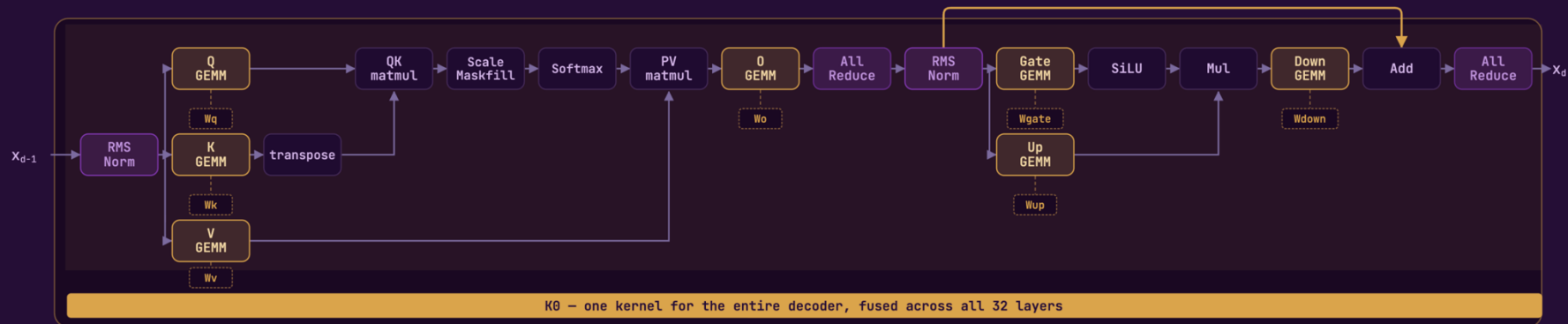
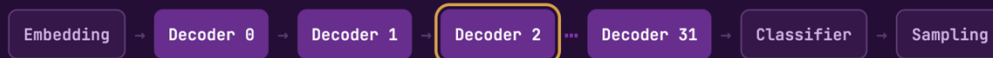


Low operator fusion

Low data locality

High launch and synchronization overheads

SN40L RDU Fuses the Entire Decoder into One Kernel



High operator fusion: one kernel call for all decoders

Zero kernel launch overheads

High data locality

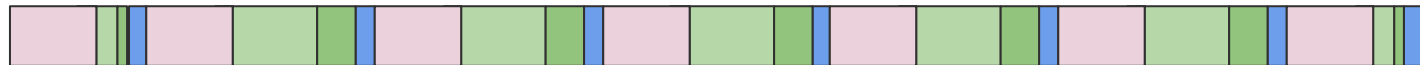
Executing Transformer on RDU

Example: Llama3.1 8B

Baseline: **100** tokens/s



+ Single Section Decoder

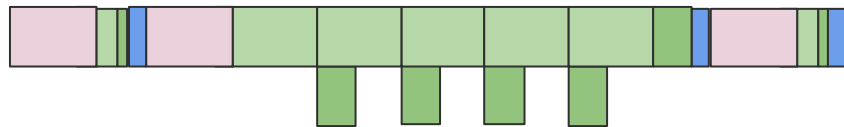


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500 tokens/s
16 SN40L

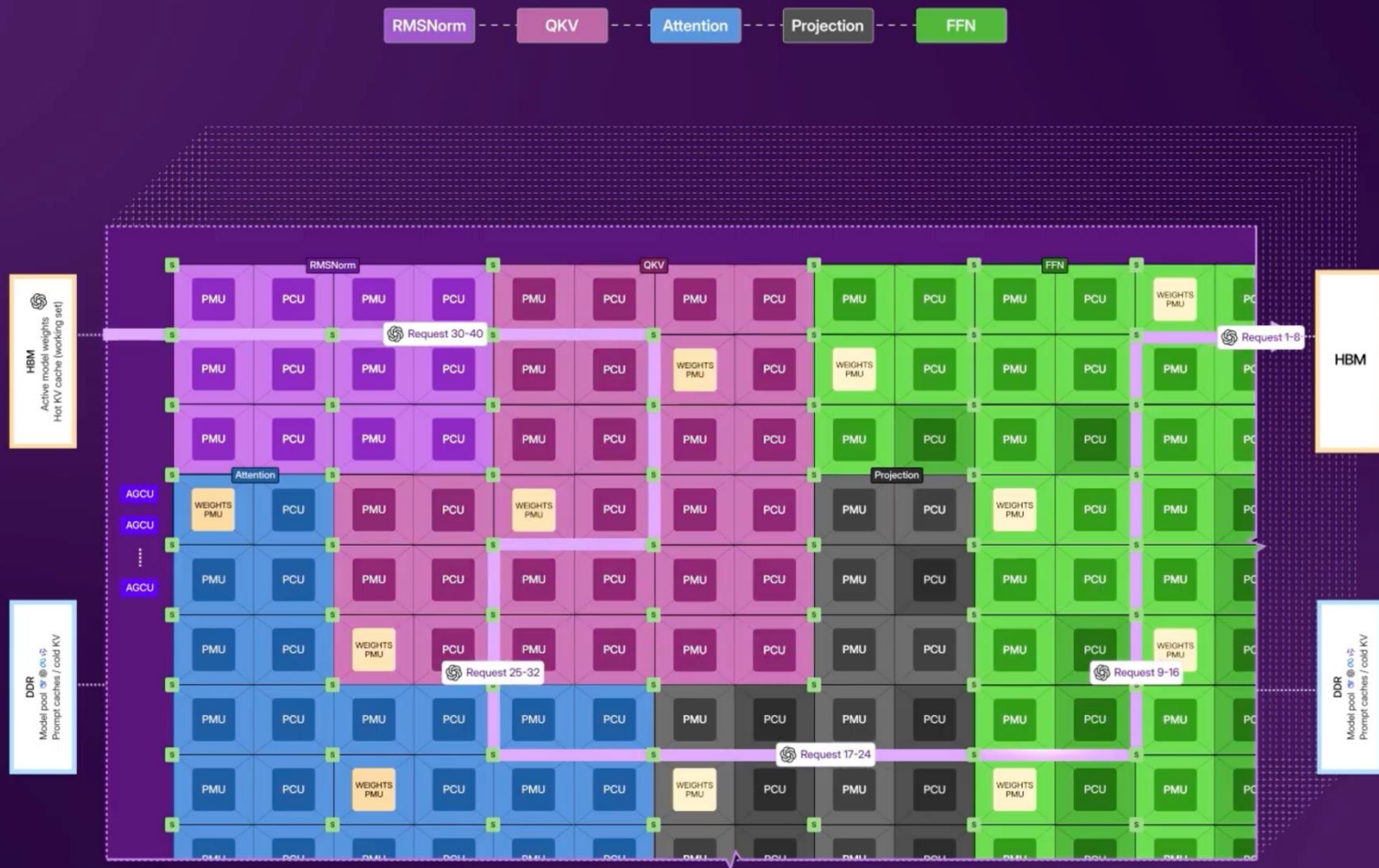


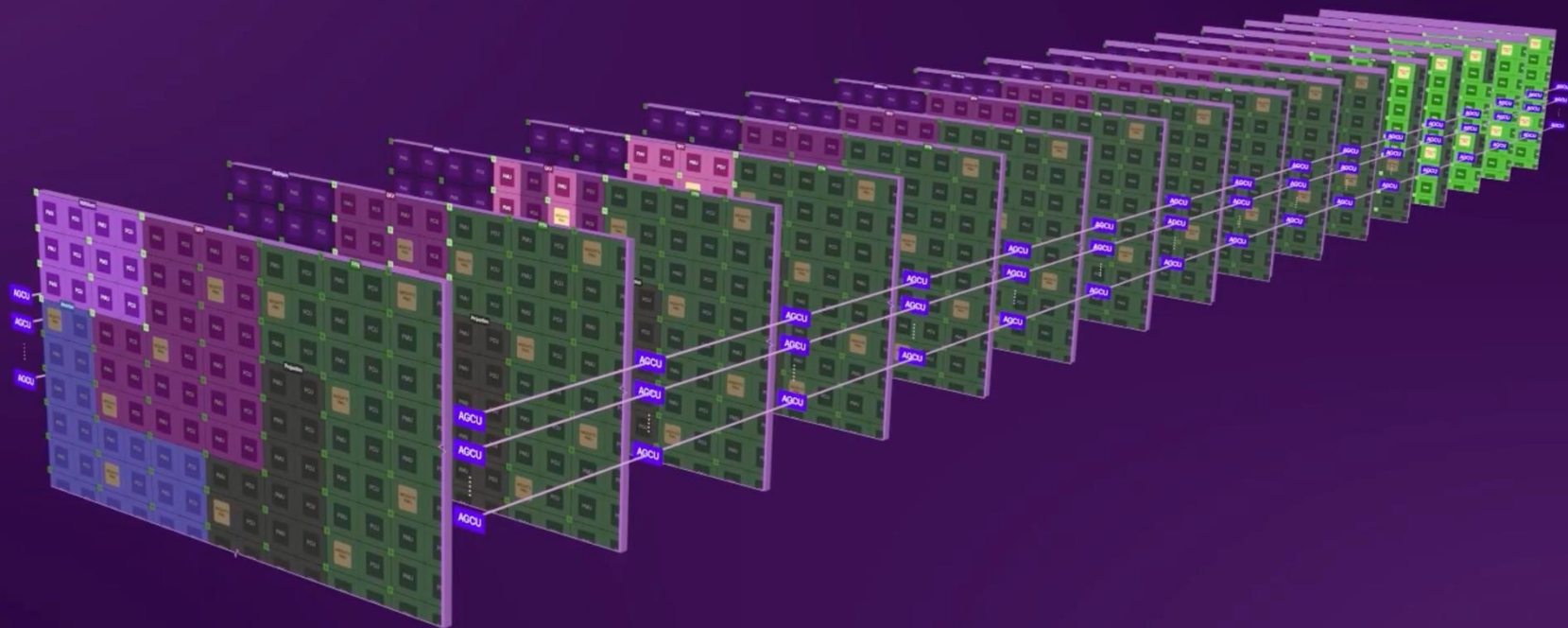
+ Kernel Looping



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1170 tokens/s
16 SN40L





16 RDUs operate as one unified system to run large models efficiently.

Lower Model Spend, Faster Output

A blended model workflow can reduce total bill while improving speed, without forcing teams to abandon the models they already rely on.

Without SambaNova

A frontier-model-heavy coding workflow can deliver strong quality, but total spend rises quickly and response speed becomes harder to scale.



Most requests go to Claude

Monthly Model Bill (Example)

~\$200K+
Higher Spend

Coding Response Speed



Slower Output



50% Lower Bill

Same outcomes, Smarter mix



3x Faster

Higher velocity for developers



Preferred Models Still in Workflow

Claude remains. Teams don't have to give it up.

With SambaNova

A blended workflow keeps high-value frontier models in the stack, while routing the right work more efficiently to improve speed and reduce total cost.



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Intelligent Routing & Inference



High-value, complex work



Fast efficient throughput

Monthly Model Bill (Example)

~50%
Lower Spend

Coding Response Speed



3x Faster Output

Takeaway: A smarter inference mix improves economics and speed without requiring a full workflow change.

Agentic Coding Demo Placeholder

Heterogeneous Disaggregated Inference



Prefill - compute dominates

Optimized for tensor parallelism and high throughput compute.

Protects TTFT.



Decode - memory bandwidth dominates

Optimized for long-context, high-batch sizes.
Either high memory bandwidth or architectures which reduce the need for memory bandwidth.

Protects TPOT.

Solving the Compute Crisis, Freeing GPU Capacity

Move high-speed decode off GPUs. Free GPU racks for prefill, training, and other compute-heavy work.

Compute Constrained Data Centers

Everything runs on GPUs

All workloads compete for the same constrained supply.

• GPU RACKS

Decode dominant load

Training compute-heavy

Prefill parallel

Other workloads misc

Decode steals capacity from everything else.

THE SHIFT

SAMBANOVA RDU



Offload decode

SambaNova Customers

01

RDU handles decode

Built for token generation. Lower per token throughput.

Fast Inference

Higher Margins

Existing Footprint

02

GPUs are freed up

No more competing capacity. GPUs return to compute-heavy where they excel

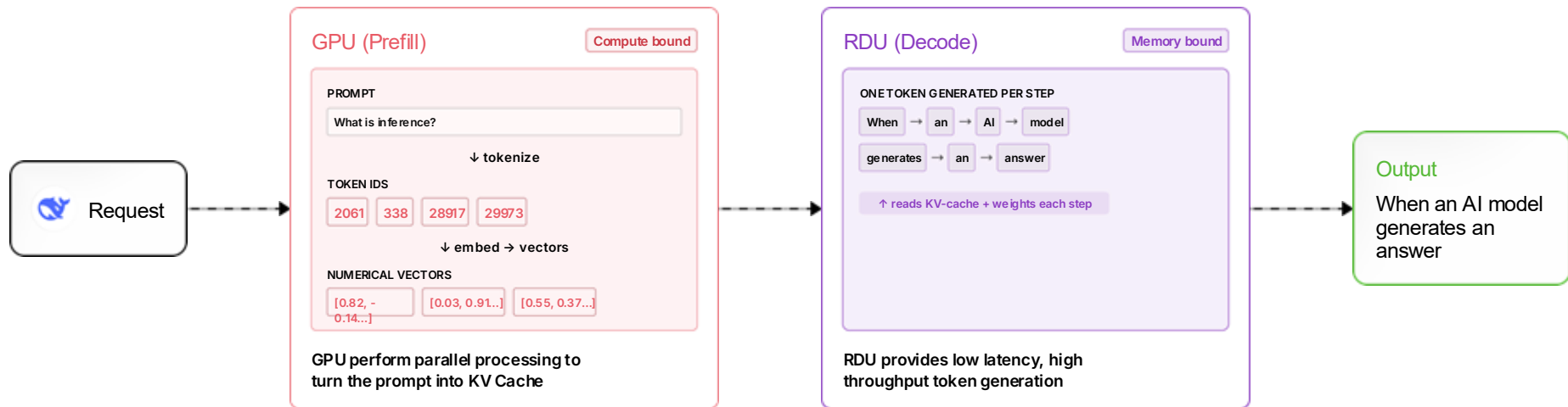
Prefill

Training

Other compute

The Building Blocks of the Agentic Inference Token Factory

Inference efficiency depends on two key metrics: how fast each request completes and how many users the system can serve.



Inference performance: NVIDIA B200 vs B200 + SN40 disaggregation

Mean end-to-end latency per agent trajectory, normalized to 33k output tokens (seconds, lower is better)

B200

8×B200 (TP=8), vLLM 0.20.1



310 s

decode 122 tok/s
prefill 4.8 s

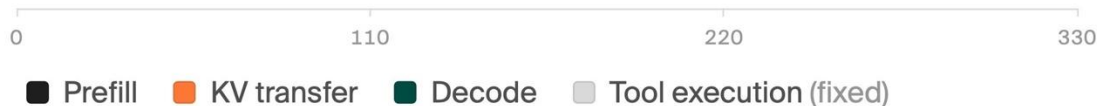
Disaggregated

B200 (TP=4) prefill + SN40 decode



162 s

decode 356 tok/s
prefill 16.5 s



Single-stream, MiniMax-M2.5. Each bar is the mean of 5 trajectories (×5 runs) = prefill + KV-transfer + decode + a fixed 34.6 s tool-execution band, with decode at N = 33,000 output tokens — MiniMax-M2.7's measured per-trajectory output on GDPval-AA. M2.5 and M2.7 share the same architecture, so decode speed carries over; only output volume is scaled. Standalone B200 measured similarly at TP=4 and TP=8.

SN50: Purpose-Built for Fast Decode

SambaNova helps providers serve agents with less infrastructure using existing data centers.

- Premium Fast Tokens
- Energy Efficiency
- Agentic Caching

SambaNova Partners with CSPs to Serve Agents with Less Infrastructure using Existing Data Centers

SN50 RDU

5nm TSMC

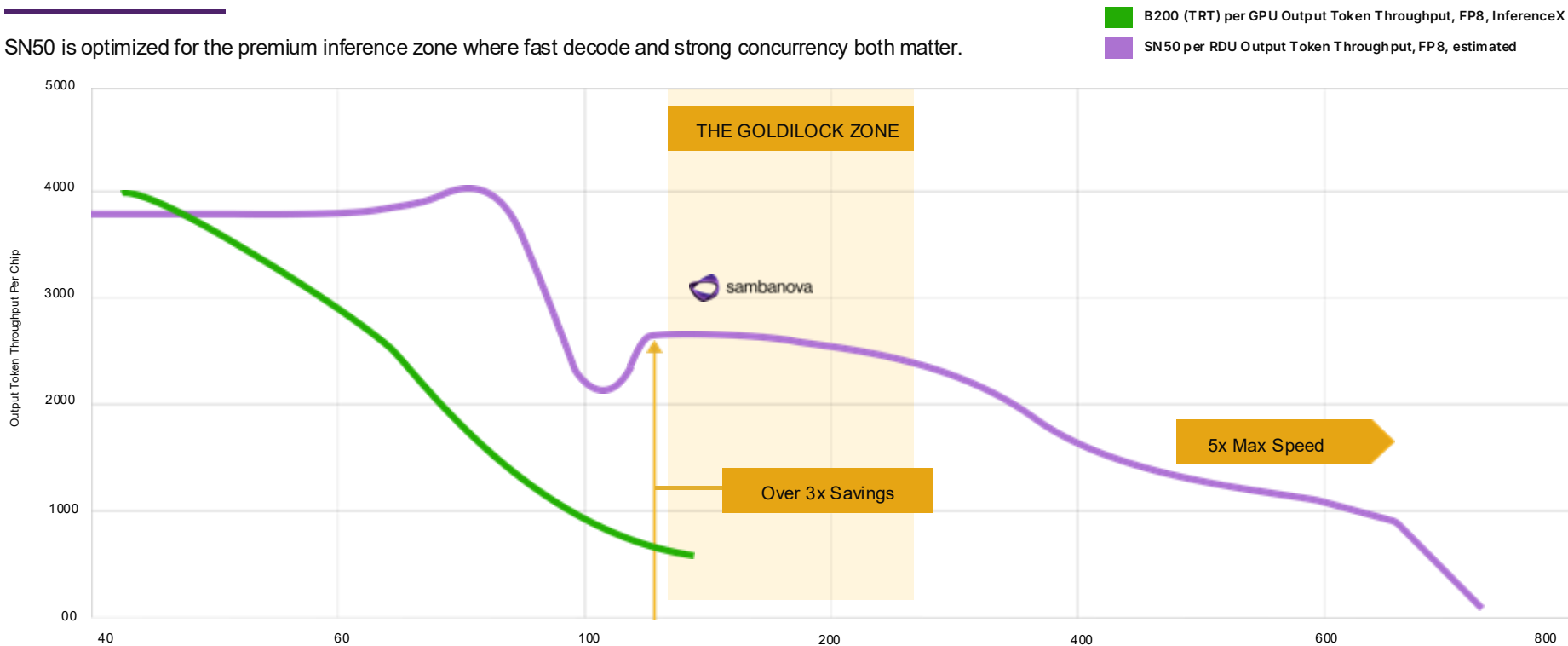
Reconfigurable Dataflow Architecture

Three-Tier Memory (SRAM, HBM_e2, DDR)



The Best Balance of User Speed and System Throughput

SN50 is optimized for the premium inference zone where fast decode and strong concurrency both matter.



Where to go from here

Try it with your own engineering team. The plugin is open source; SambaNova Cloud is the fastest path to MiniMax tokens; regional partners offer SambaNova-hosted infrastructure if you'd rather buy locally.

SambaNova Cloud

Direct access to MiniMax 2.7 and other open-source models on purpose-built inference.

cloud.sambanova.ai

Sovereign AI partners

Regionally hosted and owned Sambanova Hardware based inference services

Infercom — Europe

cloud.infercom.ai

Argyll Data Systems — UK

argylldev.com

Southern Cross — Australia

scx.ai

